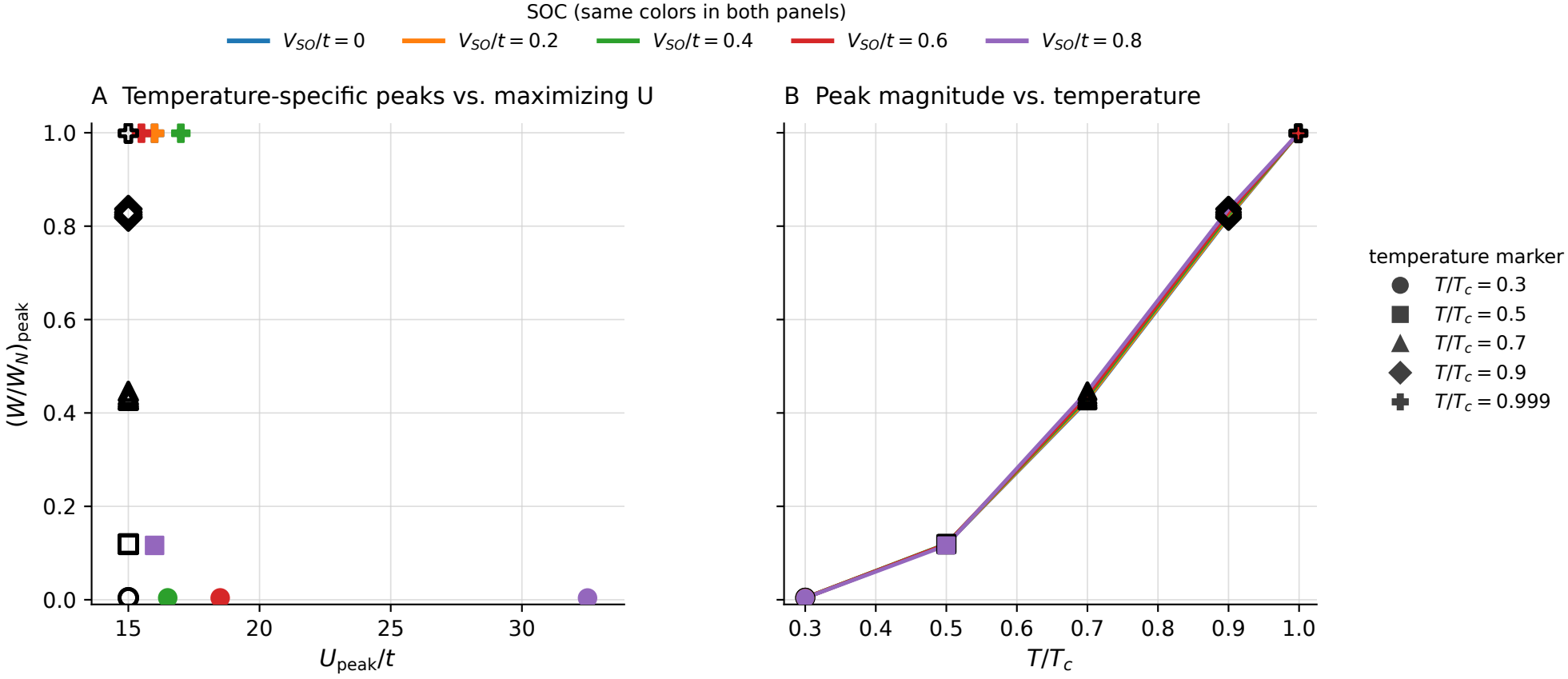

Relaxed roots; out-of-band points are diagnostics, not strict physical acceptance.

U sampled at 0.5t; no interpolation/smoothing; 17 sampled-U endpoint peak(s).